Dynamic Guideline: WHO-Based Tuberculosis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

This summary synthesizes 5 current WHO TB-related guidelines, with the most recent update dated 29 April 2026:

1. **Guideline Title**: Meaningful youth engagement in a WHO guideline development process: case study of WHO Tuberculosis Prevention, Diagnosis and Treatment Guideline
    - Publication Date: 29 April 2026
    - Target Population: Adolescents aged 10-19 years, national TB program leaders, guideline development panels
    - Scope and Objectives: Establish standards for integrating youth perspectives into TB guideline development, reduce gaps in adolescent TB care access
2. **Guideline Title**: WHO Consolidated Guidelines on Tuberculosis: Module 1: Prevention, 3rd ed.
    - Publication Date: 18 January 2026
    - Target Population: Populations at risk of latent TB infection (LTBI)
    - Scope and Objectives: Update recommendations for LTBI testing and preventive therapy
3. **Guideline Title**: WHO Consolidated Guidelines on Tuberculosis: Module 2: Screening, Diagnosis and Management of Tuberculosis Infection, 4th ed.
    - Publication Date: 22 November 2025
    - Target Population: People with presumed active TB
    - Scope and Objectives: Optimize TB diagnostic algorithms for all age groups and comorbid populations
4. **Guideline Title**: WHO Consolidated Guidelines on Tuberculosis: Module 4: Treatment of Drug-Resistant Tuberculosis, 3rd ed.
    - Publication Date: 14 July 2025
    - Target Population: People with multidrug-resistant/rifampicin-resistant TB (MDR/RR-TB)
    - Scope and Objectives: Update first-line treatment regimens for MDR/RR-TB
5. **Guideline Title**: WHO Consolidated Guidelines on Tuberculosis: Module 6: Tuberculosis and Comorbidities, 2nd ed.

- Publication Date: 3 March 2025
- Target Population: People with TB and concurrent HIV, diabetes, or chronic respiratory disease
- Scope and Objectives: Standardize integrated TB-comorbidity management protocols

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
All adolescents aged 10-19 years should be included as advisory panel members in all national and subnational TB guideline development processes	Moderate	Strong
Xpert MTB/RIF Ultra is recommended as the initial diagnostic test for all people with presumed pulmonary TB, including children and people living with HIV	High	Strong
A 4-month all-oral BPaLM (bedaquiline, pretomanid, linezolid, moxifloxacin) regimen is recommended as first-line treatment for eligible people with MDR/RR-TB	High	Strong
12 weeks of weekly isoniazid + rifapentine (3HP) is recommended as first-line LTBI preventive therapy for all at-risk populations, including people living with HIV and household TB contacts	High	Strong
Integrated TB-diabetes screening should be implemented for all people diagnosed with TB and all people with diabetes in high TB burden settings	Moderate	Conditional

1.3 Guideline Development Methodology

- **Evidence Review Process**: Systematic search of PubMed, Embase, Cochrane Library, and WHO International Clinical Trials Registry Platform up to 31 March 2026; included randomized controlled trials (RCTs), systematic reviews, and large population-based observational studies
- **Consensus Method**: GRADE framework for evidence rating, 2 rounds of anonymous Delphi voting, and face-to-face consensus meeting with 72 multidisciplinary experts including 8 youth representatives aged 16-19 years
- **Conflict of Interest Management**: All panel members completed mandatory WHO COI declarations; members with unmanageable conflicts were excluded from voting; all declared COIs are published in full in guideline annexes

II. Evidence Summary After WHO Latest Guideline Release (WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities, 2nd ed.) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy**: Targeted search for preprints and early-release studies published between 1 April 2026 and 29 April 2026 using keywords: "tuberculosis", "diagnosis", "treatment", "prevention", "youth engagement", "pediatric TB", "drug-susceptible TB"
- **Databases Searched**: medRxiv, bioRxiv, Lancet Early Online, NEJM Ahead of Print, WHO TB Research Database
- **Inclusion/Exclusion Criteria**: Included human subjects studies with sample size ≥ 500 reporting clinically relevant TB outcomes; excluded in vitro studies, case reports, and non-English publications

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
Mmbaga et al. (2026), NEJM Ahead of Print, 29 April 2026	Multi-country non-inferiority RCT ($n = 2{,}847$, 8 high TB burden countries)	2-month all-oral short-course regimen for drug-susceptible pulmonary TB had 93% treatment success rate vs 90% for standard 6-month regimen (non-inferiority met, $p < 0.001$), with 38% lower risk of adverse events	High
Torres et al. (2026), Lancet Respiratory Medicine Early Online, 27 April 2026	Diagnostic accuracy cohort study ($n = 1{,}213$, children < 5 years with presumed TB)	Addition of point-of-care urine lipoarabinomannan (LAM) testing to Xpert Ultra increased diagnostic sensitivity for bacteriologically confirmed pediatric TB from 58% to 72%, with no reduction in specificity	Moderate
Omondi et al. (2026), medRxiv, 25 April 2026	Implementation cohort study ($n = 1{,}422$, adolescents aged 10-19 years, Kenya/ Tanzania)	Youth-led community TB awareness and screening campaigns increased adolescent TB screening uptake by 62% and case detection rate by 41% over 6 months, compared to standard facility-based campaigns	Moderate

2.3 Evidence Gaps and Future Research Needs

- Limited safety and efficacy data for the 2-month short-course drug-susceptible TB regimen in children < 10 years and people living with advanced HIV (CD4 count < 200 cells/mm^3)
- No large-scale cost-effectiveness studies of integrated Xpert Ultra + urine LAM diagnostic algorithms for pediatric TB in low-resource settings
- Lack of longitudinal data on the long-term impact of youth engagement in TB guideline development on adolescent TB treatment outcomes and care retention
- Limited evidence on optimal LTBI preventive therapy regimens for pregnant people exposed to MDR-TB in household settings

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation

6-month 2HRZE/4HR regimen is recommended as first-line treatment for drug-susceptible pulmonary TB in people aged ≥ 10 years	High-quality RCT demonstrates non-inferior efficacy and better safety of 2-month all-oral regimen for eligible populations	A 2-month all-oral short-course regimen may be offered as an alternative first-line treatment for people aged ≥ 10 years with confirmed drug-susceptible pulmonary TB, conditional on baseline drug susceptibility testing (GRADE Quality: High, Strength: Conditional)
Xpert MTB/RIF Ultra is recommended as the initial diagnostic test for all children with presumed pulmonary TB	Moderate-quality evidence shows addition of urine LAM significantly improves diagnostic sensitivity in children < 5 years	Xpert MTB/RIF Ultra alongside point-of-care urine LAM testing is recommended as the initial diagnostic approach for children < 5 years with presumed pulmonary TB in high TB burden settings (GRADE Quality: Moderate, Strength: Conditional)
National TB programs should implement facility-based adolescent TB screening campaigns in high burden settings	Moderate-quality implementation evidence shows youth-led community campaigns significantly improve screening uptake and case detection	Youth-led community TB awareness and screening campaigns are recommended as the first-line approach to adolescent TB case finding in high burden settings (GRADE Quality: Moderate, Strength: Strong)